

CONFIDENTIAL — TECHNICAL EVALUATION DOCUMENT — v1.0

ATF Bridge Validation Final Report

Agent Trust Fabric ↔ VeriSigil VGS Bridge

ISSUED BY OMNIX QUANTUM LTD	ISSUED TO VeriSigil AI	REPORT DATE 22 May 2026
REPORT VERSION v1.0 — Final	ATF SPEC VERSION v1.4	CLASSIFICATION CONFIDENTIAL

4 TRACES VALIDATED	0 VIOLATIONS	9 INVARIANTS PASS	3 BRIDGE MAPPINGS CONFIRMED	✓ BRIDGE VALID
--------------------------	-----------------	-------------------------	--------------------------------------	-------------------

Overall Verdict: BRIDGE VALID — FULL COMPLIANCE CONFIRMED

This report presents the results of the ATF ↔ VGS bridge validation conducted across two batches (4 traces total). All traces were generated from the live OMNIX governance pipeline, signed using Dilithium-3 (ML-DSA-65, FIPS 204 draft), and submitted for independent validation by VeriSigil AI.

All nine ATF invariants under test returned PASS status. All three bridge mapping paths (ADMITTED→ALLOW, REJECTED→DENY, MONITORING→MONITOR+oversight) were confirmed functional. Zero violations were recorded across all traces.

1. SCOPE & OBJECTIVES

The validation exercise was scoped to confirm that the VeriSigil VGS bridge correctly interprets and maps ATF governance records (DR, TAR, RCR) to VGS trust decisions. The exercise covered:

- Nominal execution path — ADMITTED TAR, NOMINAL RCR, ALLOW decision
- Rejection path — REJECTED TAR (expired DR), DENY decision, execution blocked
- Continuity stress path — ADMITTED TAR, MONITORING RCR (CES below threshold), MONITOR + oversight flag
- Multi-hop delegation — Depth-2 chain, ATF-INV-001 (MAR) at every level, budget decrement verified

All traces were generated from live OMNIX infrastructure using the ATF Field Specification v1.4. All records are PQC-signed with Dilithium-3.

2. ATF INVARIANT COVERAGE

INVARIANT	DESCRIPTION	TRACES TESTED	RESULT
ATF-INV-001 Minimum Authority Record (MAR)	Every agent action must be backed by a valid DR. chain_root_id consistent at all delegation depths.	T1, T2, T3, T4	✓ PASS
ATF-INV-002 Temporal Admissibility Gate	TAR admission requires DR to be ACTIVE at execution_ns. Expired DR → REJECTED TAR.	T1, T2, T3, T4	✓ PASS
ATF-INV-003 Authority Budget Constraint	Granted budget ≤ delegator budget. Budget decrements correctly through chain.	T1, T3, T4	✓ PASS
ATF-INV-004 Chain Root Consistency	chain_root_id must be identical across all records in a delegation chain.	T1, T2, T3, T4	✓ PASS
ATF-INV-005 Delegation Depth Monotonicity	delegation_depth must increment by exactly 1 at each sub-delegation level.	T4	✓ PASS
ATF-INV-006 PQC Signature Integrity	All DR and TAR records must carry a valid Dilithium-3 signature over content_hash.	T1, T2, T3, T4	✓ PASS
RGC-INV-002 CES Composite Score Validity	CES = weighted average of temporal, budget, context, integrity components. Score matches claimed values.	T1, T3, T4	✓ PASS
RGC-INV-004 Continuity Status Threshold	NOMINAL ≥ 80.0, MONITORING 50–79.99, CRITICAL < 50. Status reflects CES score.	T1, T3, T4	✓ PASS
RGC-INV-006 Escalation Event on MONITORING	MONITORING RCR must carry an escalation_event_id. Null escalation_event_id is a violation.	T3	✓ PASS

All 9 invariants under test returned PASS. Zero violations recorded.

3. BRIDGE MAPPING CONFIRMATION MATRIX

ATF TAR STATUS	ATF RCR STATUS	EXPECTED VGS DECISION	CONFIRMED
ADMITTED	NOMINAL	ALLOW	✓ Traces T1, T4
REJECTED	— (blocked)	DENY	✓ Trace T2
ADMITTED	MONITORING	MONITOR + oversight flag	✓ Trace T3

All three bridge mapping paths confirmed functional. Rejection mapping and continuity stress mapping validated independently.

4. TRACE-LEVEL VALIDATION DETAILS

TRACE T1 · Batch 1 · NOMINAL Path		BRIDGE: ALLOW
DELEGATION ID (DR) ATFDR-9587338A8B9147A5	TAR ID ATFTAR-09710B7B25D84E90	
RCR ID ATFRCR-5A1F6244DCA340ED	CHAIN ROOT ID ATFDR-9587338A8B9147A5	
AGENT ID AID-GOVERNANCE-a3f7c1e92b4d8f05	TASK ACTION decision_review:production:evaluate_risk_signal	
TAR ADMISSION STATUS ADMITTED	RCR CONTINUITY STATUS NOMINAL	
DR CONTENT HASH 91599b3f6a2c68bafaf0627c2c73e3fa2570b19888f0ac3c03522f7bf461e76e	TAR CONTENT HASH e0c2414e16c958632697d4f42c0da3b5f492861edb7a0b693dda82ae1e6d2940	
AUTHORITY BUDGET GRANTED 75.0 / 100.0	BUDGET REMAINING AT RCR 72.0 (–3.0 consumed)	
PQC ALGORITHM Dilithium-3 (ML-DSA-65, FIPS 204)	DR EXPIRES AT 2026-05-23T04:46:36 UTC	
CES SCORE BREAKDOWN		
Composite (CES)		98.20
Temporal		99.99
Budget		96.00
Context		97.00
Integrity		100.00
Validation Result: DR ACTIVE at TAR admission. TAR ADMITTED. RCR NOMINAL (CES 98.20 ≥ 80.0 threshold). Context drift 3.0% — within bounds. Active anomalies: 0. Sample reason: SCHEDULED. Bridge correctly mapped to ALLOW. ATF-INV-001 through ATF-INV-006 satisfied. RGC-INV-002 and RGC-INV-004 satisfied.		

TRACE T2 · Batch 2 · REJECTED TAR Path		BRIDGE: DENY
DELEGATION ID (DR) ATFDR-F2D7D54129944698 CHAIN ROOT ID ATFDR-F2D7D54129944698 TASK ACTION risk_signal_review:production:evaluate DR CONTENT HASH 45c5cbd0148296043264600583cf7bfe0505893f4797a436343a2bcab13be38c DR EXPIRED AT (NS) 1779468166437371904 NANOSECONDS AFTER EXPIRY 7,200,965,213,798 ns (≈ 2.00 hours)	TAR ID ATFTAR-ECB62496CA1349D5 AGENT ID AID-GOVERNANCE-b1e4f2a03c7d9e11 TAR ADMISSION STATUS REJECTED TAR CONTENT HASH 6780d10d16395d8a36fb8b75e8cc8e20ce76e3ff9490a30010be1b5d091a75a9 EXECUTION AT (NS) 1779475367402585702 RCR ISSUED — (execution blocked, no RCR)	
Rejection Reason: DR expired at nanosecond 1779468166437371904. Execution attempt at nanosecond 1779475367402585702 — 7,200,965,213,798 ns (approximately 2.00 hours) after DR expiry. ATF-INV-002 (Temporal Admissibility Gate) enforced. Execution blocked. No RCR issued. Bridge correctly mapped to DENY. Governance action: block execution.		

TRACE T3 · Batch 2 · MONITORING RCR — Continuity Stress Path

BRIDGE: MONITOR + OVERSIGHT

DELEGATION ID (DR)

ATFDR-945ABBB62A484DD3

RCR ID

ATFRRCR-40CEED678CBF4A39

AGENT ID

AID-GOVERNANCE-c2f5a3b04d8e0f12

TAR ADMISSION STATUS

ADMITTED

BUDGET AT ADMISSION

80.0

CONTEXT DRIFT

38.0% (above nominal threshold)

SAMPLE REASON

THRESHOLD_BREACH

TAR ID

ATFTAR-9FDF2F90350C46DE

ESCALATION EVENT ID

ATFCEE-C73F2E0C0C8F4722

TASK ACTION

compliance_check:production:evaluate

RCR CONTINUITY STATUS

MONITORING

BUDGET REMAINING

25.0 (–55.0 consumed, 68.75% fragmented)

ACTIVE ANOMALIES

2

RCR CONTENT HASH

9bfa971f9ca2b030783bf96666b774b5
c8dcf6154c63646513d8b8a37febe4b3

CES SCORE BREAKDOWN

Composite (CES)

67.77

Temporal

99.99

Budget

31.25

Context

62.00

Integrity

80.00

Validation Result:

TAR ADMITTED (DR ACTIVE). RCR MONITORING — CES 67.77 is between 50.0 and 79.99 (MONITORING band per RGC-INV-004). Budget consumption and context drift both above nominal bounds triggered THRESHOLD_BREACH sample. Escalation event ATFCEE-C73F2E0C0C8F4722 present (RGC-INV-006 satisfied). Execution survivable. Bridge correctly mapped to **MONITOR + oversight flag**.

TRACE T4 · Batch 2 · Delegation Chain Depth 2

BRIDGE: ALLOW

DR ROOT (DEPTH 1)

ATFDR-CFA91576BBEF4DEA

TAR ID

ATFTAR-0583E16B5DC54447

CHAIN ROOT ID (ALL RECORDS)

ATFDR-CFA91576BBEF4DEA

AGENT → SUB-AGENT (DEPTH 2)

AID-GOVERNANCE-d3a6b4c05e9f1023 → AID-GOVERNANCE-e4b7c5d16f0a2134

TAR ADMISSION STATUS

ADMITTED

DR ROOT CONTENT HASH

78491a081295d6e66fcb99a40c3017e1
46ae115949bd80821f5cc4d58d8595f1

DR SUB-DELEGATE (DEPTH 2)

ATFDR-3BA3E5231BC74C0F

RCR ID

ATFRRCR-9BE7EDF89F144A39

DELEGATOR → AGENT (DEPTH 1)

OPERATOR-harold-nunes-omnix → AID-GOVERNANCE-d3a6b4c05e9f1023

BUDGET DECREMENT CHAIN

100.0 → 85.0 (depth 1) → 50.0 (depth 2)

RCR CONTINUITY STATUS

NOMINAL

DR SUB CONTENT HASH

6f2b4b5448ab2176d4b7056bae1b62f9
434e9fb852c28a1bdf289ec0298c7a99

CES SCORE BREAKDOWN (DEPTH-2 AGENT)

Composite (CES)		96.60
Temporal		99.99
Budget		90.00
Context		98.00
Integrity		100.00

Validation Result: ATF-INV-001 (MAR) satisfied at both delegation levels — depth-1 and depth-2 agents each backed by valid DRs. ATF-INV-004 confirmed — chain_root_id ATFDR-CFA91576BBEF4DEA consistent across DR-root, DR-sub, TAR, and RCR. ATF-INV-005 confirmed — delegation_depth increments 1→2. ATF-INV-003 confirmed — budget decrements correctly: 100→85→50. RCR NOMINAL (CES 96.60). Bridge correctly mapped to **ALLOW**.

FORMAL ATTESTATION — OMNIX QUANTUM LTD

I, Harold Alberto Nunes Rodelo, Founder & Chief Executive Officer of OMNIX QUANTUM LTD, hereby attest that:

- 1. All four traces presented in this report were generated from the live OMNIX governance pipeline on 22 May 2026.
- 2. All Delegation Receipts (DR) and Temporal Admissibility Records (TAR) are signed with Dilithium-3 (ML-DSA-65, FIPS 204 draft). Signatures are available in the raw JSON source files and may be independently verified against the OMNIX public key previously provided to VeriSigil AI.
- 3. All field values, content hashes, and CES scores presented in this report are unmodified from the pipeline output.
- 4. This report constitutes the final output of the ATF ↔ VGS Bridge Validation — Batch 1 and Batch 2. The bridge is confirmed valid. No further batches are required to establish invariant compliance.

This report is issued under the terms of the VeriSigil AI ↔ ATF/OMNIX Integration Scope Memo and Mutual NDA, both executed 22 May 2026.

Harold Alberto Nunes Rodelo
Founder & Chief Executive Officer, OMNIX
QUANTUM LTD

Date: 22 May 2026